SPORTS & ARTHROSCOPY · NEET-SS / FET / INI-SS

Hip arthroscopy and FAI

1. Femoroacetabular impingement (FAI): concept

- **Abnormal contact between the proximal femur and acetabular rim** during motion (especially **flexion + internal rotation**): **labral tears and chondral damage**: a **cause of early OA** in young adults (Ganz).
- **Warwick Agreement (2016) definition of FAI syndrome: triad of symptoms (motion- or position-related hip/groin pain), clinical signs (positive impingement test, restricted internal rotation) and imaging findings (cam or pincer morphology)**. **Morphology alone is not FAI** (common in asymptomatic athletes).

2. Types

	Cam	Pincer
Pathology	**Femoral**: **aspherical femoral head-neck junction** (reduced offset, "pistol-grip" deformity); jammed into the acetabulum in flexion/IR	**Acetabular**: **global or focal over-coverage** (retroversion, coxa profunda, protrusio)
Typical patient	**Young, athletic males** (adolescent sports during physeal closure: **repetitive loading of the developing physis**); **SCFE sequelae**, Perthes	**Middle-aged active women**

	Cam	Pincer
Damage pattern	**"Outside-in" abrasion**: **chondrolabral separation**, **delamination of the anterosuperior acetabular cartilage** (**carpet lesion/wave sign**): **more cartilage damage**	**Labral crushing/degeneration first** (linear contact); **contrecoup lesion posteroinferiorly** (head levered out); ossification of the labrum
Measures	**Alpha angle (over 55 degrees)** (Notzli; on 45-degree Dunn view or radial MRI), **head-neck offset ratio (under 0.17)**, **pistol-grip deformity**, **anterior offset**	**Crossover sign** (retroversion: anterior wall crosses the posterior wall), **posterior wall sign** (posterior wall medial to the head centre), **ischial spine sign** (projection of the ischial spine into the pelvis: retroversion), **lateral centre-edge angle (LCEA) over 40 degrees**, **coxa profunda (acetabular fossa medial to the ilioischial line)**, **protrusio (femoral head medial to ilioischial line)**, **Tönnis angle (acetabular index) under 0 degrees**

- **Mixed (combined) FAI: commonest (about 70-85%)**.
- **Subspine (AIIS) impingement:** prominent **anterior inferior iliac spine** (Hetsroni types I-III) (**healed rectus femoris avulsion**): **AIIS decompression**.
- **Ischiofemoral impingement:** narrowing between the ischium and lesser trochanter (**quadratus femoris oedema** on MRI).
- **Femoral version abnormalities: femoral retroversion** worsens impingement; **excessive femoral anteversion/acetabular dysplasia**: **instability, not impingement**: hip arthroscopy alone may worsen symptoms.

3. Clinical evaluation

- **Groin pain ("C sign": hand cupped over the lateral hip)**, worse with **sitting, flexion, pivoting**, mechanical symptoms (catching, clicking), **reduced internal rotation in flexion**.
- **Tests:**
  - **FADIR (anterior impingement test: flexion, adduction, internal rotation)**: **highly sensitive, low specificity**,
  - **FABER (Patrick)**: posterior/SI or posterior impingement,
  - **Posterior impingement test** (extension + external rotation),
  - **Log roll** (instability/intra-articular),
  - **Dial test** (capsular laxity),
  - **Stinchfield (resisted straight leg raise)**: intra-articular pathology.
- **Diagnostic intra-articular local anaesthetic injection**: pain relief confirms an intra-articular source.

4. Imaging

- **AP pelvis (standardised: coccyx tip about 1-3 cm above the symphysis, centred)**: **crossover sign, posterior wall sign, ischial spine sign, LCEA, Tönnis angle, joint space (Tönnis OA grade)**, **pistol-grip deformity**.
- **45-degree Dunn view / frog-leg lateral / cross-table lateral**: **alpha angle, head-neck offset**.
- **False profile view (Lequesne)**: **anterior centre-edge angle (ACEA under 20 degrees = anterior undercoverage)**, AIIS.
- **MR arthrography (gadolinium) or 3T MRI** (labral tears: **anterosuperior** commonest, cartilage damage), **CT with 3D reconstruction** (cam/pincer mapping, **femoral version**, AIIS).

Acetabular dysplasia and instability measures (important to exclude)

- **LCEA (Wiberg): normal 25-40 degrees**; **under 20 degrees = dysplasia**; **20-25 degrees = borderline dysplasia**.
- **Tönnis (acetabular) angle: normal 0-10 degrees**; **over 10 degrees = dysplasia (steep roof)**; **under 0 degrees = over-coverage (pincer)**.
- **Femoral head extrusion index** (over 25% = uncovered), **Shenton line break**, **FEAR index** (femoro-epiphyseal acetabular roof index: helps decide instability in borderline dysplasia).
- **Dysplasia (LCEA under 20) treatment = periacetabular osteotomy (PAO; Bernese/Ganz)**: **isolated arthroscopic labral debridement/capsulotomy in dysplasia can cause catastrophic instability**.

5. Indications and contraindications for FAI surgery

- **Indications: symptomatic FAI syndrome** failing non-operative management (**activity modification, physiotherapy (core and hip strengthening), NSAIDs, injection**) for at least about 3 months, with **preserved joint space**.
- **Contraindications / poor prognostic factors: established OA (Tönnis grade 2 or more, joint space under 2 mm)**: **poor outcomes; conversion to THA**, **older age (over 40-50)**, **significant dysplasia (LCEA under 20)** (unless combined with PAO), **severe chondral damage (Outerbridge IV)**, **high BMI**, workers' compensation.
- **Evidence: UK FASHIoN trial** (and **Australian FAIT**): **hip arthroscopy improved patient-reported outcomes more than best conservative care (physiotherapy)** at 12 months in FAI syndrome.

6. Surgical technique

- **Supine or lateral position on a traction table**, **well-padded perineal post** (or **post-less**), **traction (about 8-10 mm of joint distraction)**; **limit traction time to under about 2 hours**

(**pudendal nerve neurapraxia** (perineal post), **sciatic nerve**, **lateral femoral cutaneous nerve** (anterior portal), **perineal skin injury**, **scrotal/labial injury**).

- **Portals:**
  - **Anterolateral (first; under fluoroscopy; 1 cm anterior and 1 cm superior to the greater trochanter tip)**: **superior gluteal nerve** (about 4-6 cm superior),
  - **Mid-anterior / anterior**: **lateral femoral cutaneous nerve (closest structure: about 3 mm)**, **femoral neurovascular bundle** (medial: avoid crossing the line from the ASIS downward),
  - **Posterolateral**: **sciatic nerve** (keep the leg in neutral/internal rotation).
- **Central compartment** (with traction): labrum, chondral surfaces, ligamentum teres, acetabular fossa; **peripheral compartment** (without traction, hip flexed): **head-neck junction (cam)**, synovium, loose bodies.
- **Interportal or T-capsulotomy**; **capsular repair/closure** (**reduces post-op instability**, especially in **borderline dysplasia, hyperlaxity, women**).
- **Pincer: acetabular rim trimming (acetabuloplasty)** with **labral takedown and re-fixation** (**avoid over-resection: iatrogenic instability/dysplasia**; LCEA should remain over 25 degrees).
- **Labrum: repair/refixation (suture anchors) preferred over debridement** (better outcomes; **labral seal effect**: maintains intra-articular fluid pressure, cartilage nutrition, stability); **labral reconstruction (allograft/ITB autograft)** for irreparable/deficient labra in young patients.
- **Cam: femoroplasty (osteochondroplasty)** of the head-neck junction under fluoroscopy and dynamic assessment; **avoid over-resection (over about 30% of the neck diameter: femoral neck fracture risk)**; **protect the retinacular (lateral epiphyseal) vessels** (**posterosuperior neck: branches of the medial femoral**

circumflex artery: **avoid resection at the 12 o'clock-posterior region**: **AVN**).

- **Chondral lesions**: **microfracture** for small full-thickness acetabular lesions.
- **Open surgical hip dislocation (Ganz: trochanteric flip osteotomy, preserving the MFCA by staying anterior to the short external rotators; the obturator externus protects the deep branch)**: for **complex/global deformities (severe pincer, protrusio, posterior cam, Perthes deformity, SCFE)**.
- **Periacetabular osteotomy (PAO)** for dysplasia (± arthroscopy); **femoral derotation osteotomy** for abnormal version.

7. Complications of hip arthroscopy

- **Neurapraxia** (**pudendal (perineal post: commonest neurological), sciatic, femoral, LFCN**), **traction injuries**, **iatrogenic chondral/labral damage** (scuffing on portal entry), **instrument breakage**.
- **Heterotopic ossification** (**NSAID prophylaxis (naproxen) after femoroplasty**).
- **Iatrogenic instability/dislocation** (unrepaired capsulotomy, over-resection, dysplasia, hyperlaxity).
- **Femoral neck fracture** (over-resection), **AVN** (rare).
- **Abdominal fluid extravasation** (**intra-abdominal compartment syndrome**; risk with iliopsoas release, high pump pressure, prolonged surgery).
- **DVT**, **infection** (rare), **under-resection** (**residual impingement: commonest cause of revision**), **adhesions**.

8. Other hip arthroscopy indications and extra-articular conditions

- **Ligamentum teres tears** (instability, dysplasia), **loose bodies** (synovial chondromatosis, trauma), **septic arthritis** (washout), **pigmented villonodular synovitis (diffuse TGCT)**, **gluteus medius/minimus tears ("rotator cuff of the hip")**: **endoscopic or open repair**; **greater trochanteric pain syndrome** (gluteal tendinopathy: **education + loading exercise (LEAP trial) best long-term**; ESWT; steroid injection short-term).
- **Snapping hip:**
  - **External (ITB over the greater trochanter)**: **visible snap**; ITB release/Z-plasty,
  - **Internal (iliopsoas tendon over the iliopectineal eminence or femoral head)**: **audible snap**, **extension from a flexed-abducted-externally rotated position**; **iliopsoas release (at the lesser trochanter or transcapsular)** if refractory (weakness of hip flexion),
  - **Intra-articular** (labral tears, loose bodies).
- **Deep gluteal syndrome** (sciatic nerve entrapment: **piriformis**, fibrous bands): endoscopic decompression.
- **Athletic pubalgia ("sports hernia"/core muscle injury)** and **adductor-related groin pain** (**Doha agreement** classification: adductor-, iliopsoas-, inguinal-, pubic-related groin pain); **osteitis pubis**.
- **Femoral neck stress fractures: compression side (inferomedial: protected weight bearing; fix if fracture line over 50% of the neck width)** vs **tension side (superolateral: urgent internal fixation)** (**female athlete triad/RED-S**).

Exam pearl: **Cam (young men, alpha angle over 55, outside-in chondral delamination) vs pincer (middle-aged women, crossover/posterior wall/ischial spine signs, LCEA over 40, contrecoup). Mixed commonest. FADIR sensitive.**

Exam pearl: **Arthroscopy contraindicated with joint space under 2 mm/Tönnis 2+ and in dysplasia (LCEA under 20: needs PAO). Repair the labrum (seal) and the capsule. Avoid over-resection (neck fracture) and posterosuperior retinacular vessels (AVN). Under-resection = commonest cause of revision. FASHIoN: arthroscopy beats physio.**

Exam pearl: **Portals: anterior portal = LFCN closest; anterolateral = superior gluteal nerve; posterolateral = sciatic nerve. Perineal post = pudendal neurapraxia. HO prophylaxis with NSAIDs. Fluid extravasation into the abdomen = abdominal compartment syndrome.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.